



Deep learning-based forgery identification and localization in videos

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Abstract

The video forensics capabilities are constantly improving in terms of evidence accumulating, analysis, processing, and storage. Video forensic analysis involves scientific investigation, comparison, and/or assessment of video files that are considered as proof in the court. In this paper, we focus on inter-frame video forgery detection and localization with respect to frame inserting and deleting that are essential from digital video forensics perspective. We design a 3-dimensional convolutional neural network (3DCNN) model for detecting video inter-frame forgery and localize forgery using multi-scale structural similarity index measurement algorithm. We introduce an absolute difference algorithm to differentiate video frames from each other which minimize the temporal redundancy and identify forgery artefacts within the video frames. This improves the 3DCNN efficiency and accuracy in detecting frame insertion and deletion forgery. The proposed model outperforms existing models in terms of accuracy, precision, recall, and F1 score in various post-processing operations, compression rates, and video length.

Keywords Digital video forgery · Inter-frame video forgery · Frame insertion · Frame deletion · Video forgery localization

1 Introduction

In the fast progressing era of digital technology, video forgery has become an easy and on-going task for the users of smart devices. Videos that are manipulated affect the authenticity and integrity of the video files because the statistical characteristics of the videos are modified and abnormalities are introduced. The statistical analysis when deeply carried out ensures easy identification of the artefacts or footprints left during the tampering process. Digital video forensics is introduced to ensure the authenticity of the videos. Video tampering is classified into: 1. temporal tampering, 2. spatial tampering, and 3. spatial-temporal tampering. The video tampering that occurs in the spatial or in the spatial-temporal domain is categorized as intra-frame video forgery, and the video tampering that occurs in the temporal domain is categorized as inter-frame video forgery. The localization of frame insertion and deletion plays an important role in analysing

forged video files during cyber-crime analysis. Inter-frame video forgery distorts the sequence of frames in four different ways: 1. frame insertion, 2. frame deletion, 3. frame duplication, and 4. frame shuffling [14]. Our work focuses on detecting and localizing video inter-frame forgeries (inserting and deleting) since frame replication and shuffling do not play a major role in *cybercrime*. In the literature, there are various techniques proposed for the detection of inter-frame video forgeries [1]. Most of the techniques are based on manual feature extraction to identify the artefacts of forgery, and these features are sensitive to post-processing operations (light inconsistencies, noise, compression, blurring, etc.). In addition, many of the techniques are unable to perform efficiently and cannot be used for digital video forensics as they are trained and tested on same datasets. Hence, the performance of the models in detecting inter-frame video forgeries is affected.

Deep learning encourages researchers to use the models of machine learning and deep learning to investigate the forgeries in the field of digital video forensics. It has been used in various application domains like action recognition, image classification, and object recognition. The two most popular deep learning algorithms that perform well in pattern recognition of images are convolutional neural network (CNN) and recurrent neural network (RNN).

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We propose a 3D convolutional neural network (3DCNN) model to overcome the drawbacks of existing methods and identify video inter-frame forgeries for which the original source is not available. In addition, we use inherent temporal abnormalities to locate the amount of tampered video frames based on the temporal disparity between neighbouring video frames of the genuine video clip and its recreated clip. The multi-scale structural similarity (MS-SSIM) index measurement analysis algorithm is used to localize video inter-frame forgery (inserting and deleting).

The remaining part of the paper is organized as follows: Sect. 2 discusses the related review of inter-frame video forgery detection and localization. Challenges and contributions are discussed in Sect. 3. Section 4 describes the proposed model. The results and analysis are presented in Sect. 5. Section 6 highlights the conclusion and future work.

2 Related work

Many authors have earlier contributed significant methods in digital video forensics for detecting digital inter-frame video forgery. We divide these methods in terms of feature engineering-based and deep learning-based. From the feature engineering-based methods, Stamm et al. [16] presented an approach for detecting frame deletion or insertion operated by an increase in motion prediction error in P-frame and used this method to design an anti-forensic technique on forged videos. The model only works with fixed group of pictures (GOP) sizes. Wang et al. [17] proposed optical flow analysis to authenticate and identify inter-frame forgeries in digital videos (insertion, deletion and duplication). The technique is robust to some degree on MPEG compression but limited due to the computational burden.

Inter-frame forgery detection in MPEG-2 and H.264 encoding is proposed by Kingara et al. [7] by utilizing the footprints from motion, brightness gradient features, optical flow coefficients, and prediction residual. The performance of the method suffers when used on videos with strong light source, dynamic videos and variable GOPs.

The feature engineering-based analysis methods work well on the small number of frames in a video and suffer when the number of frames in the video is more. In addition, these methods are computationally complex and consume a lot of duration in processing, affecting the performance evaluation metrics.

The deep-learning-based methods, particularly CNNs, have exceptionally achieved better results in many computer vision problems, particularly in large-scale image classification. Recently, efficient CNN models are used for detecting inter-frame forgeries in videos. Long et al. [9] was first to use the CNN model for detecting frame duplication forgery using a coarse-to-fine deep convolutional neural network. When a

video contains multiple sequences of duplicated frames, the model's performance is limited. Kaur and Jindal [6] proposed a highly efficient method to detect an inter-frame forgery in videos using in-depth CNN that utilizes spatial and temporal correlation between the frames. The model showed the identification of abnormalities within the frames. CNN models using a transfer learning approach were suggested by Nguyen et al. [11] for detecting video forgeries in inter-frame but was unable to extract the temporal features accurately. The authors presented a video inter-frame forgery dataset (VIFFD) that includes insertion, deletion, and duplication forgeries for experimental analysis.

Fadl et al. [3] utilised a 2D convolutional neural network (2D CNN) and structural similarity (SSIM) measurement to identify inter-frame forgeries in videos (insertion, deletion, and duplication). The classification is done with a Gaussian RBF multi-class support vector machine (RBF-MSVM). The computational cost of using 2DCNN for detecting inter-frame forgery is very high, and no cross-dataset evaluation was performed on unknown dataset by the proposed model. Bakas et al. [2] presented a forensic approach based on prediction footprint variation (PFV) to detect inter-frame forgery in surveillance footage and locate forgery using a range of motion vectors. When one or more GOPs are deleted or inserted in a video sequence, the method fails.

3 Challenges and contributions

The major challenge is to identify efficient features from a particular dataset. Traditional methods evaluating forged videos (high quality and lengthy) have demonstrated poor performance and are low efficient. We use deep learning-based 3DCNN model [5] to extract high-dimensional features from video files since 3DCNN model is more efficient in extracting spatial and temporal features. The temporal correlations across consecutive frames should be carefully examined to improve video forensics analysis. The main contributions of the paper are:

1. To learn high-dimensional features from input video frames, we design an 3DCNN model considering Conv3D layers.
2. Unlike the normal 3DCNN model, we propose an absolute difference algorithm to evaluate the difference of successive frames that minimize temporal redundancy among the frames and expose the artefacts of forgery operation in videos.
3. Localization of inter-frame forgery in video (insertion and deletion) is carried out by spatial-temporal analysis using multi-scale structural similarity (MS-SSIM) index measurement algorithm.

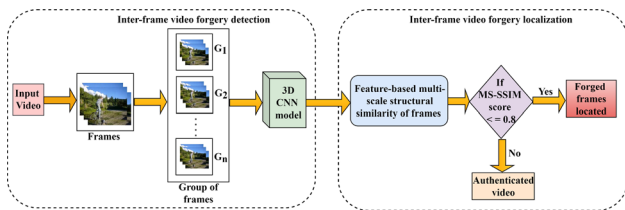


Fig. 1 Proposed model for detecting video inter-frame forgery and localization

4. Creation of inter-frame video forgery (insertion and deletion) dataset using UCF-101 action classes.

4 Proposed methodology

Hand-crafted feature extraction techniques used for the detection of inter-frame video forgery result in limited solution, computational complexity, and dataset processing limitations. Due to these constraints, the performance of the model is affected using hand-crafted feature extraction techniques. DL-based 3DCNN technique is used for extracting prominent features from training datasets solving various problems of computer vision. We propose a 3DCNN model for detecting inter-frame video forgery since 3DCNN are capable of generating spatio-temporal features more accurately. The model proposed is divided into inter-frame video forgery detection and inter-frame video forgery localization as shown in Fig. 1.

4.1 Inter-frame video forgery detection

In this section, we divide the process of inter-frame video forgery detection into video frame pre-processing and 3DCNN model.

4.1.1 Video frame pre-processing

A video is composed of a series of sequential image frames that appear to be identical to one another. To differentiate the video frames from each other, an absolute difference layer is added at the beginning of the 3D CNN model. The equation 1 computes the pixel-wise discrepancy between frame f and the adjacent frame $f + 1$. The output deviation of the frame heap is fed into the 3DCNN model.

$$P_f(m, n) = |K_f(m, n) - K_{f+1}(m, n)|$$

$$1 \leq m \leq w, 1 \leq n \leq h$$

$$D_k(m, n) = \begin{cases} 1, & \text{if } P_f(m, n) > f \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

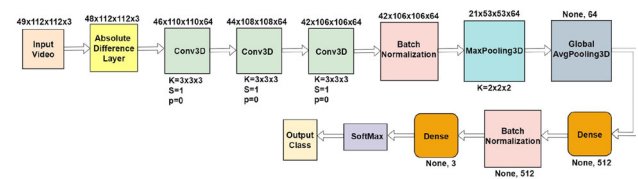


Fig. 2 The proposed 3DCNN for detecting video inter-frame forgery

$P_f(m, n)$ is the difference frame, $K_f(m, n)$ is the intensity of pixel (m, n) in the f th frame, and w and h are the width and height of the video frames. During the process of inter-frame video tampering, some discernible artefacts obligatorily exist and to detect such artefacts, we evaluate the absolute disparity among the sequential frames for minimizing the temporal redundancy. In addition to absolute difference layer, the video frames are pre-processed to construct a group-of-frames to deal with large video files to improve accuracy and reduce computations. There are 49 frames in each group to improve performance in the proposed model. The group size depends on the computational resources which may vary. Consider an input video V_1 consisting of F frames with dimensions $W \times H$ pixels. The video frames are divided into equal group-of-frames G each of length P frames. Each group of frames G is represented by a number of 3D frames that depict the stream of video. Equation 2 is mathematically represented as:

$$V_1 = \bigcup_{p=1}^P G_p \quad (2)$$

G_p is the p th group-of-frames G , and P is the overall frames of the input video.

4.1.2 3DCNN model

Network architecture Three convolutional three-dimensional (Conv3D) layers are considered in the proposed 3DCNN for detecting video inter-frame forgery followed by rectified linear units (ReLU), batch normalization (BN), MaxPooling3D, Global Average Pooling, Dense, SoftMax and output class as shown in Fig. 2. We use DL-based 3DCNN for detecting inter-frame video forgery, since they inherently apply the convolutions in the 3D space and are best suitable for extracting features from spatial and temporal domain. Inter-frame forgery attacks are performed in the temporal domains. The Conv3D kernel sizes are stated as *no. of kernels* $\times$ (*height* $\times$ *width* $\times$ *inputs*). The feature maps size among the layers is given as *no. of featuremaps* $\times$ (*height* $\times$ *width*). To maintain the shape of frames, we apply padding in all the layers. To aggregate the max/mean of a certain feature over a specific region, we use the MaxPooling operation. The global average pooling (GAP) layer is being used to accomplish

an even more dimensionality reduction, in which the GAP layer lowers each $h \times w$ feature map to a single value by calculating the average of $h \times w$ values. To ensure the performance of the output activations, the batch normalisation method is used [4]. It also helps 3DCNN handle overfitting issues while achieving a well-balanced generalisation as part of the regularisation process. After several successive fully connected layers, a softmax layer for multi-class classification is employed to produce the desired output. With the aid of fully connected layers and a softmax layer, the feature vectors are transformed into 3D probabilities. Finally, the results classified are shown based on the 3D probabilities.

4.2 Inter-frame video forgery localization

We use multi-scale structural similarity to localize the inter-frame video forgery, which is discussed in the following sections:

4.2.1 Multi-scale structural similarity index measurement

Multi-scale structural similarity (MS-SSIM) index measurement is used to examine the dissimilarity of video frames and to measure the duration of video inter-frame forgery (insertion and deletion). Multiple-scale modelling of visual intensity (luminance), contrast, and structure serves as the basis for the MS-SSIM [10]. The primary goal of MS-SSIM is to assess the quality of video frames and to achieve improved detection accuracy while processing in real time. The MS-SSIM starts the process of locating inter-frame video forgery by using a referential frame and comparing with the succeeding frames. The succeeding frame is then considered as a reference frame, and the procedure is repeated with the next succeeding frames. The MS-SSIM achieves better accuracy and faster processing speed of the normal structural similarity (SSIM) index measure. The structural similarity assessment is a measure for comparing two video frames' structural similarity. The method attempts to measure the visibility of defects (discrepancies) between an inconsistent and a reference image in order to evaluate perceptual image quality. The structural similarity (SSIM) [18] extracts three main features from a video frame: luminance, contrast and structure. The SSIM score is given by:

$$\text{SSIM}(x, y) = [L(x, y)]^\alpha \cdot [C(x, y)]^\beta \cdot [S(x, y)]^\gamma \quad (3)$$

where α , β , and γ define the weight given to each model of luminance $L(x, y)$, contrast $C(x, y)$, and structure $S(x, y)$.

4.2.2 MS-SSIM algorithm

1. Consider only one frame at a time f_t .
2. At time $t+1$, acquire the next frame f_{t+1} .

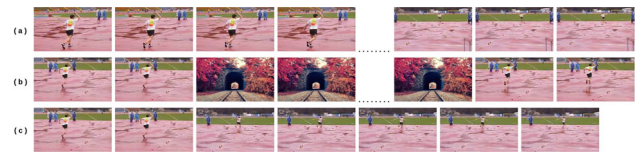


Fig. 3 UCF-101 sample video showing **a** authentic, **b** insertion forgery, and **c** deletion forgery

3. Calculate the multi-scale structural similarity of two consecutive frames f_t and f_{t+1} .
4. A frame forgery is identified with reference to insertion or deletion if the similarity value is below the pre-determined threshold $\theta = 0.8$ between two consecutive frames. As per the algorithm, a change in the threshold value beyond 0.8 will not drastically affect the accuracy rate of localizing frame insertion or frame deletion, but a threshold value below 0.8 will affect the accuracy rate of localization.

5 Result and analysis

5.1 Datasets

The proposed model is evaluated on UCF-101 [15] and VIFFD [12] datasets. There are currently no standard datasets dedicated to video inter-frame forgeries. So, by picking some action videos from the UCF-101 dataset, we generated insertion and deletion forgery videos. For our experiments, we selected 700 videos of different actions and generated inter-frame forgery (insertion and deletion) using ffmpeg tool.

In the creation of frame insertion and deletion forgery videos, we varied the number of frames from 10 to 150 frames. The selected videos are compressed with H.264 and MPEG-4 using libraries libx264 and libavcodec of the ffmpeg tool with an adaptive group-of-pictures (AGOP) structure. The videos have frame rates ranging from 25 to 30 frames per second. We chose 60 videos and applied post-processing operations such as Gaussian blurring, Gaussian noise, and brightness variations to analyse inter-frame video forgery detection under different conditions.

The VIFFD dataset has 392 videos with a combination of authentic, insertion, deletion and duplication videos collected from five surveillance cameras of real life scene shots with different lightning conditions. For the experimental analysis, we chose 30 authentic videos, 30 insertion forgery videos, and 30 deletion forgery videos. The inter-frame video forgery is differentiated into multi-class classification, namely insertion, deletion and authentic with a total of 2190 videos. Figure 3 shows the sample video from UCF-101 dataset showing authentic and forged (insertion and deletion) video.

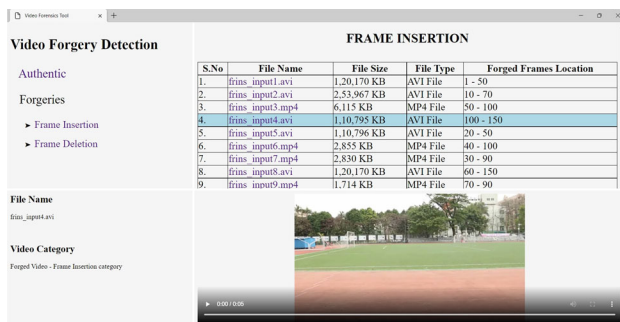


Fig. 4 GUI-based sample application showing video forensic analysis

5.2 Implementation details

To carry out our implementation, we utilized 8GB GPU (NVIDIA RTX 2080) with Intel i7, running at 2.20GHz processor and 32 GB RAM. We used Python3.0, OpenCV, ffmpeg, Keras 2.6.0 with Tensorflow 2.6.0 framework to build high-level deep learning model.

GUI-based implementation Figure 4 depicts a GUI-based sample application that will benefit video forensic investigators in analysing video files. The application's user interface consists of three sections (left, right and bottom). All video forgery categories (Authentic and Forgeries) are presented in the left section in a tree view. With the detection of forgery, the video files will be placed in the appropriate category. When we select a category in the left section, the right section will display all of the videos in that category, along with meta-data. The bottom section is used to play the video for analysing the content of a video forgery listed under the right section.

5.3 Experimental analysis

The evaluation criteria and experimental analysis of our model are discussed in this section. To evaluate the proposed model's performance, we consider detection accuracy (Acc), which is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

where TP—true positive, the amount of accurately identified video forgeries is known as true positive; FP—false positive, the amount of incorrectly identified video forgeries is known as false positive; TN—true negative, the amount of accurately identified original videos is known as true negative, and FN—false negative, the amount of incorrectly identified original videos is known as false negative. The model's effectiveness is satisfactory when detection of precision rate, recall rate

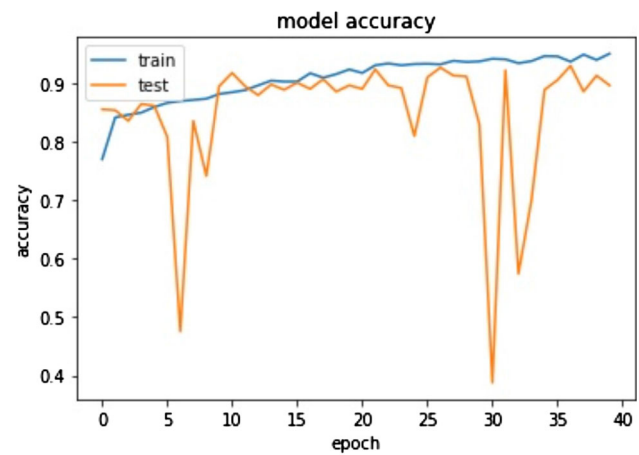


Fig. 5 Train versus test accuracy

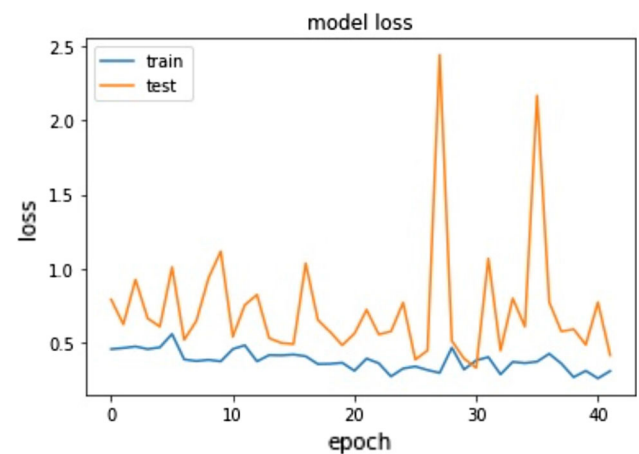


Fig. 6 Train versus test loss

and F1 score are achieved better.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$F1 = 2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$$

The network starts the learning process with a learning rate of 0.001 and weight decay of 0.0005. Adam (adaptive moment estimation) is used as the loss function to optimize the 3DCNN-based model. We use VIFFD as a testing dataset to validate the trained model's performance. The pre-processing for testing dataset is the same as the training dataset. The input to the trained 3DCNN model is the test dataset, and the class labels for each video clip are obtained; if any one of the class is predicted with insertion or deletion forgery, then the frame is marked as tampered frame. The video frame is authentic when the class is predicted with authentic.

Table 1 Performance evaluation showing predictive test on VIFFD dataset

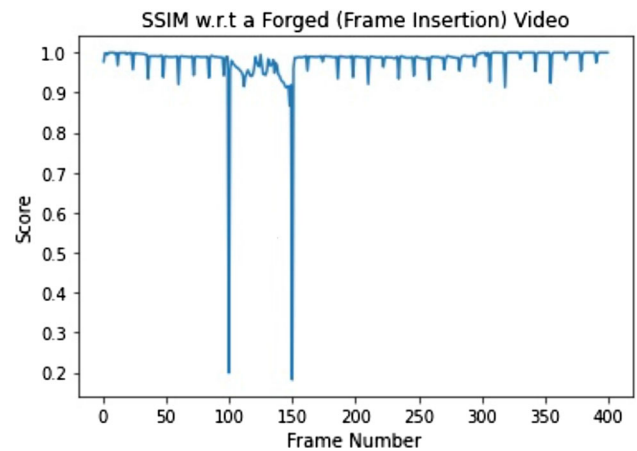
	Precision	Recall	F1-score	Support
FrameDeletion	1.0	0.92	0.96	75
FrameInsertion	0.95	1.00	0.98	119
Authentic	1.00	1.00	1.00	124
Accuracy			0.98	318
Macro avg.	0.98	0.97	0.98	318
Weighted avg.	0.98	0.98	0.98	318

Support—indicates the total number of groups in frame insertion, frame deletion, and authentic videos

Training and testing results with UCF-101 dataset The proposed 3DCNN model is evaluated on UCF-101 dataset which is split randomly in a ratio of 75:25 for training and testing, with batch size of 16, and number of epochs carried out is 40. The categorical cross-entropy is utilized as a loss function to assess the efficiency of a multi-class classification model with probability values as its output. The resulting evaluation metric used is accuracy, precision, recall and F1 score. Figures 5 and 6 show accuracy and loss results after training and testing the proposed model on UCF-101 dataset. The graph depicts the variance in accuracy and loss. After 40 epochs, the loss begins to decrease as the learning improves. The proposed model's training progress reaches its highest level of precision. The proposed model is robust to video forgery in static scene and dynamic scene with multiple forgeries. Once the model is trained, we tested and predicted the model with VIFFD dataset. The predicted evaluation matrix gives a recovery implication about the errors that our trained model is making and examines the performance of the classification process. The predicted rate of evaluation matrix in Table 1 shows predictive test on VIFFD dataset.

It is evident from these results (Table 2) that the proposed model by its learning capability can accurately differentiate inter-frame forgery and authentic videos.

Video insertion forgery localization Every video frame is matched with the additional frames temporally and spatially during the localization of insertion forgery in a video, and the MS-SSIM score is calculated. If the MS-SSIM score of the

**Fig. 7** Frame insertion localization based on MS-SSIM

frames in the inserted position is substantially smaller than the actual value, there will be two discontinuities in insertion forgery with a dropping peak value where the functionalities of consecutive frames differ significantly. In Fig. 7, the visual output results in frame insertion video forgery where the frames 100–150 are inserted from other videos which are indicated by two falling peak values. The values of similarity in the locations where the frames are inserted show significantly lower than the normal values. There will be two distinct declining peak values for inter-frame insertion forgeries which indicate that the frames within the dropping peak values are from different videos.

Video deletion forgery localization In frame deletion forgery, more video frames are deleted to cover up the truth in the videos. Two neighbouring frames in an authentic video will have a high degree of similarity. But, the similarity of the frames for a long period may suddenly change, and the score value will become lower at the deleted position. The MS-SSIM score results less immediately in frame deletion video forgery resulting in a single falling peak value. In Fig. 8, the inter-frame deletion video forgery results with single falling peak value where 50 frames are deleted (75–125 frames). The frame deletion operation results in a greater interval between two similar frames, resulting in a dropping peak value.

Table 2 Performance evaluation over conventional and DL methods

Method	Forgery operation	No. of videos	Accuracy	Precision	Recall	F1-score
QoCCLB [21]	Insertion/deletion	599	–	0.95	0.92	0.93
PR&MB [19]	Deletion	770	–	0.72	0.66	0.699
ZOCMCFA [8]	Insertion/deletion	60	–	0.83	0.85	–
CNN [20]	Insertion/deletion	128	0.85	–	–	–
MO-BWO+CNN [13]	Insertion/deletion	09	0.85	0.825	–	0.826
Proposed model	Insertion/deletion	2190	0.98	0.96	0.95	0.97

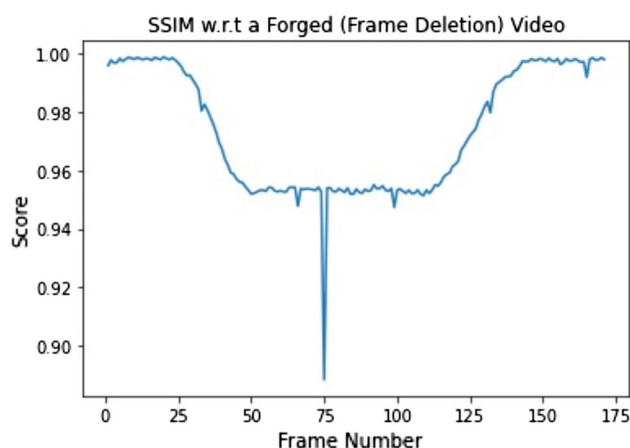


Fig. 8 Frame deletion localization based on MS-SSIM

5.4 Comparison

In accordance with detecting video inter-frame forgery, the 3DCNN-based model outperforms both conventional and other deep learning-based approaches. We compared our proposed model with QoCCLB [21], PR&MB [19], ZOCCMFA [8], CNN [20] and MO-BWO+CNN [13]. The comparison results are shown in Table 2.

The proposed method produces better results in terms of accuracy, precision, recall, and f1-score than the existing models.

Ablation study We chose 60 videos and applied post-processing operations like compression rates, varying video length, Gaussian noise, Gaussian blurring, and brightness variations. These videos were trained and tested on our proposed model with and without applying difference algorithm. The results achieved were compared with the existing methods [8,22] under insertion and deletion forgery. The accuracy of these methods is 0.7478 and 0.7724, respectively. We achieved the accuracy of 0.8234 without difference algorithm and 0.9817 with difference algorithm.

6 Conclusion

Digital video forensics plays a key role in analysing cyber-crime since digital video editing software tools make easier to tamper video files for the illegitimate benefit and produce tampered evidence in the court. In this work, we propose a 3-dimensional CNN model for detecting inter-frame forgery in video and use the MS-SSIM approach to localize the forgeries. The video forgery detection model includes an absolute difference algorithm, to minimize the temporal redundancy and identifies forgery artefacts to improve the forgery detection efficiency. The proposed 3DCNN model learns more relevant characteristics to detect video inter-frame forgeries

with high classification accuracy. The existing methods on inter-frame video forgery detection are not completely supporting the forensic investigators in detecting video forgery when no source video is available and only the forged video is provided. Our proposed model overcomes the above-mentioned challenge and outperforms when compared to the existing models in both static and dynamic videos. In the future work, we focus on distinct forms of video inter-frame forgery detection, namely frame duplication and shuffling for applications other than digital video forensics. In addition, we will design a standard inter-frame video forgery dataset that can be used by the research community for future research in the digital video forensics field.

Data availability The authors confirm that the data supporting the findings of this study are available within the article [and/or] its supplementary materials.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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